

LETTERS

TO THE EDITOR

Re: Willemink MJ, van Es HW, Helmhout PH, et al. The effects of dynamic isolated lumbar extensor training on lumbar multifidus functional cross-sectional area and functional status of patients with chronic nonspecific low back pain. *Spine* 2012;37:E1651–8.

Willemink *et al*¹ recently published results from a prospective single-arm trial examining the effects of isolated lumbar extension (ILEX) resistance training (RT) upon lumbar multifidus (MF) cross-sectional area (CSA), and disability using the Roland-Morris disability questionnaire. Investigating the effects of exercise interventions in chronic low back pain to understand mechanisms involved in producing clinical improvements is of considerable value. This is particularly important considering recent suggestions that conditioning effects of exercise interventions (*i.e.*, improved strength, endurance, or hypertrophy) are not associated with change in clinical outcomes.² However, as Willemink *et al*¹ examined a highly specific exercise, ILEX, their study could potentially provide more specific information. We have previously highlighted the importance of specificity.³ Unfortunately, the methodology of this study creates difficulty in surmising why no change in MF CSA occurred despite improved Roland-Morris disability questionnaire scores. Absence of control group notwithstanding, particular concern pertains to control of RT variables in their intervention.

Their RT intervention did not control frequency, load or effort. During the 12- to 24-week duration of training, participants only performed sessions at a frequency suiting their convenience. In addition, load was not controlled by using a relative percentage of participant's one repetition maximum or maximal voluntary contraction on the ILEX device, nor did participants train to momentary muscular failure (MMF) to control effort. Control of these factors has been highlighted recently as of considerable importance in RT research.⁴ It is reasonable to suggest the lack of change in MF CSA was because both intensity of load and effort were insufficient to induce muscular adaptation. Choi *et al*⁵ reported significant improvement in MF CSA in participants who underwent postlumbar discectomy during a 12-week ILEX RT program, however, did not provide description of RT variables. Mooney *et al*⁶ alternatively reported significant improvement in total CSA of the combined lumbar extensors after 8 weeks of ILEX RT 2 times per week with load controlled at 50% maximal voluntary contraction and effort controlled

by participants performing repetitions to MMF. Willemink *et al*¹ do however note that these results cannot offer inductive conclusions regarding the MF specifically due to use of total CSA values. Recent research shows that intensity of effort affects hypertrophy resulting from RT (*i.e.*, continuing repetitions to MMF appearing key^{7,8}), thus it is unsurprising no significant effect resulted from the intervention Willemink *et al*¹ used. Indeed, lack of strength improvement from ILEX training in previous studies that did not control intensity of load or effort further suggests this to be the case.^{4,9,10}

Considering methodological issues with this study,¹ we urge readers to consider its results carefully. It seems difficult to conclude that changes in clinical outcomes are wholly unrelated to conditioning effects of RT based interventions. Although the improvements in Roland-Morris disability questionnaire (≈ 3 points, $\approx 13\%$) were similar to improvements seen in another study (≈ 4 points, $\approx 16\%$) performing 10 weeks of ILEX RT 1 time per week, using a load permitting 6 to 12 repetitions before achieving MMF,¹¹ neither of these studies used control groups, rendering it impossible to rule out a placebo type effect.

In our opinion, it seems premature to rule out improvement in conditioning of the lumbar spine as unrelated to improvements in clinical outcomes. Psychosocial factors evidently play some prognostic role in clinical outcomes from exercise interventions including ILEX,¹¹ yet it is not clear whether they are a determinant of clinical improvement or perhaps an effect.¹² Yet the aforementioned conditioning effects may also play some role when considered specifically in isolation.³ Indeed we have just published results from a randomized controlled trial using 12 weeks of ILEX RT 1 time per week, controlling load using 80% maximal voluntary contraction, and effort by having participants perform repetitions to MMF. We reported significant moderate correlations between changes in ILEX strength and both pain measured by visual analogue scale ($r = -0.488$ to -0.668) and disability measured by Oswestry Disability Questionnaire ($r = -0.414$ to -0.539).¹³

We support research examining whether the proposed mechanisms of exercise interventions are indeed responsible for clinical improvements. However, we wish to note that room still exists for improved research to address this, and currently good reason exists to think specific exercise interventions such as ILEX are of great value in conditioning the lumbar spine and which might be responsible for improvements in clinical outcomes such as pain and disability.

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